

Project Executable Files

Date	27 June 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This project provides an intelligent agricultural production optimization system using machine learning techniques. It analyzes soil nutrients and environmental factors to recommend suitable crops. The system helps farmers improve crop yield and resource efficiency. It promotes sustainable and data-driven agricultural decision-making.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Pending
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Pending

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

OptiCrop/

```
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|
├── app.py
├── crop_prediction_model.pkl
├── requirements.txt
├── README.md
├── dataset/
|   └── crop_recommendation.csv
├── templates/
|   ├── index.html
|   ├── prediction.html
|   └── assessment.html
├── static/
|   ├── css/
|   └── images/
├── notebooks/
|   └── model_training.ipynb
├── models/
|   └── trained_model.pkl
└── tests/
    └── test_cases.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	To be updated by Team Lead
Login Credentials (Demo)	Demo account if required
Platform / Hosting Provider	Flask / Local or Cloud
Repository Link	To be updated by Team Lead
Demo Video Link	To be updated by Team Lead

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Install Python 3.x and required packages using requirements.txt

- Open terminal and navigate to project folder
- Run: `pip install -r requirements.txt`
- Start application using: `python app.py`
- Open browser and access local Flask URL

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Accuracy depends on dataset quality	Improve dataset continuously
2	Limited real-time weather integration	Future enhancement
3	Supports selected crop datasets only	Extend dataset coverage